



A species interaction kick-starts ecological speciation in allopatry

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Adaptation to different environments is thought to play a key role in speciation. However, speciation typically begins in allopatry, where reproductive isolation can also arise through neutral processes or selection unrelated to ecological differences. Disentangling the role of adaptive ecological divergence in the early stages of speciation therefore remains an important challenge in understanding the origin of new species. Here, we study threespine stickleback populations that have recently evolved in isolated postglacial lakes either in the presence or absence of prickly sculpin—a resource competitor that also shares ubiquitous trout predators with stickleback. We simulated secondary contact between several stickleback populations from these two ecological contexts in large, seminatural ponds, and genotyped offspring from 411 mating events to assess the strength of premating isolation associated with this biotic factor. Assortative mating between populations of the same ecological type (i.e., both sculpin-sympatric or both solitary) was moderate on average but ranged from weak to complete. Strikingly, and in line with a central premise of ecological speciation, the strength of premating isolation increased with increasing morphological and genomic population divergence shaped by sculpin-mediated selection. In contrast, overall phenotypic and genomic population divergence agnostic to sculpin presence/absence only poorly explained premating isolation, highlighting how ecological speciation in allopatry can be obscured by other sources of divergence. More broadly, our findings demonstrate how interactions with other ecologically similar species can play a major role in initiating and driving evolutionary trajectories toward new species, even in allopatry.

assortative mating | biotic interactions | divergent natural selection | allopatric speciation | reproductive isolation

The formation of new species typically begins with geographically isolated populations (1–6), yet what triggers speciation among allopatric populations remains uncertain. In allopatry, reproductive isolation can arise via diverse mechanisms, including neutral processes and different forms of selection (4, 7, 8). For instance, random genetic drift, intragenomic conflict, and stabilizing or uniform selection can all generate reproductive isolation in geographically isolated populations, which then becomes apparent upon their secondary contact (4, 9–16). Divergent ecological selection can also cause reproductive isolation, whereby reproductive barriers arise as a by-product of adaptation to contrasting ecological conditions (17, 18). Although ecological divergence is an evident driver of speciation with gene flow and necessary for the later coexistence of species, its significance in the initial allopatric phase of speciation remains poorly understood. Clarifying whether and how ecological divergence drives speciation in allopatry is therefore a central and unresolved challenge in speciation research.

Several issues impede efforts to understand the role and causes of ecological divergence in speciation, especially in allopatric contexts. First is the classic problem that if speciation cannot be observed directly at its onset, the barriers detected later may not be those initially key to the process (19, 20). This uncertainty makes it difficult to tell whether ecological differences between extant species—especially older ones—were essential during their incipient speciation, or whether they emerged only later (21–23). Second, measuring reproductive isolation between allopatric populations in the wild is challenging—if not impossible—because it requires testing them under simulated secondary contact. Only by removing the geographic barrier originally separating populations can the strength of reproductive isolation be quantified. Finally, attributing observed reproductive isolation to ecological divergence requires quantifying divergence in response to differences in natural selection, a task that is especially difficult among allopatric populations where other sources of variation can obscure the contribution of ecological adaptation to overall divergence. For these reasons, a powerful approach to testing whether and how ecological

Significance

New species arise as populations evolve reproductive isolation, a process that typically starts in geographic isolation (allopatry). However, whether and how ecological differences in allopatry promote speciation remains unclear. We study stickleback fish populations that have recently evolved (<12,000 y) in postglacial lakes with or without sculpin—another fish species that competes for resources and shares predators with stickleback. Using a large-scale pond experiment, we show that sculpin-driven selection has led to varying, sometimes complete premating isolation. Isolation strength is predicted by the extent of adaptive divergence in both body shape and the genome. Our findings reveal a link between ecological adaptation and the early evolution of reproductive isolation in allopatry, addressing a key gap in ecological speciation theory.

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divergence drives incipient allopatric speciation is to quantify the strength of divergence among young, replicate populations in response to a known selective pressure and then assess its association with direct measures of reproductive isolation in realistic scenarios of secondary contact.

Here we provide such a test of incipient ecological speciation in allopatric populations from contrasting biotic environments. We focus on recently evolved postglacial populations of threespine stickleback fish (*Gasterosteus aculeatus*) from small freshwater lakes in western British Columbia (Canada). Previous work has shown that these geographically isolated populations were independently founded from an ancestral marine population less than 12,000 y ago and subsequently evolved independently, adapting largely from standing genetic variation brought to freshwater during colonization (24–27) (*SI Appendix, Supplementary Analysis 1*). Our study leverages a subset of these populations from lakes that are geographically proximate and ecologically similar except for the presence or absence of a single other fish species, prickly sculpin (*Cottus asper*). As a bottom-dwelling fish from a similar ecological guild, sculpin compete with stickleback for benthic food resources but also elicit indirect selective effects on stickleback through shared (trout) predators (28, 29). Together, these interactions have repeatedly driven phenotypic and genetic divergence of stickleback from lakes with and without sculpin (26, 28–31). For example, sympatric stickleback populations co-occurring with sculpin exhibit increased planktivory and a body shape better adapted to locomotion and prey capture in the open-water (limnetic) habitat. In contrast, stickleback from lakes without sculpin are more

benthic in their diet and phenotype. Despite these overall differences, the extent of divergence along this benthic-limnetic niche axis—a key axis of fish diversification more broadly (32–34)—varies considerably among stickleback populations (26, 29, 30). Such variation in the response to a focal ecological variable is common in nature because of additional environmental sources of variation, differences in the strength of selection, variable constraints in responding to selection, stochastic events, or some combination of these (35–38).

Ecological speciation occurs when reproductive isolation evolves as a by-product of adaptation to different environments (17, 18). The natural variation in the evolutionary response of stickleback to sculpin therefore provides an opportunity to test a central prediction of ecological speciation; namely, that the degree of divergent adaptation should correspond to the strength of reproductive isolation. To this end, we combine measures of genetic and phenotypic divergence with experimental measures of reproductive isolation among several “solitary” (occurring without sculpin) and sculpin-sympatric (hereafter simply “sympatric”) stickleback populations to test whether adaptation to the presence/absence of this single interacting species has kick-started incipient speciation in allopatry.

Results

We conducted an experiment to quantify reproductive isolation among four sympatric and three solitary stickleback populations, representing a continuum of divergence associated with sculpin presence/absence (29). While clearly physically separated from one

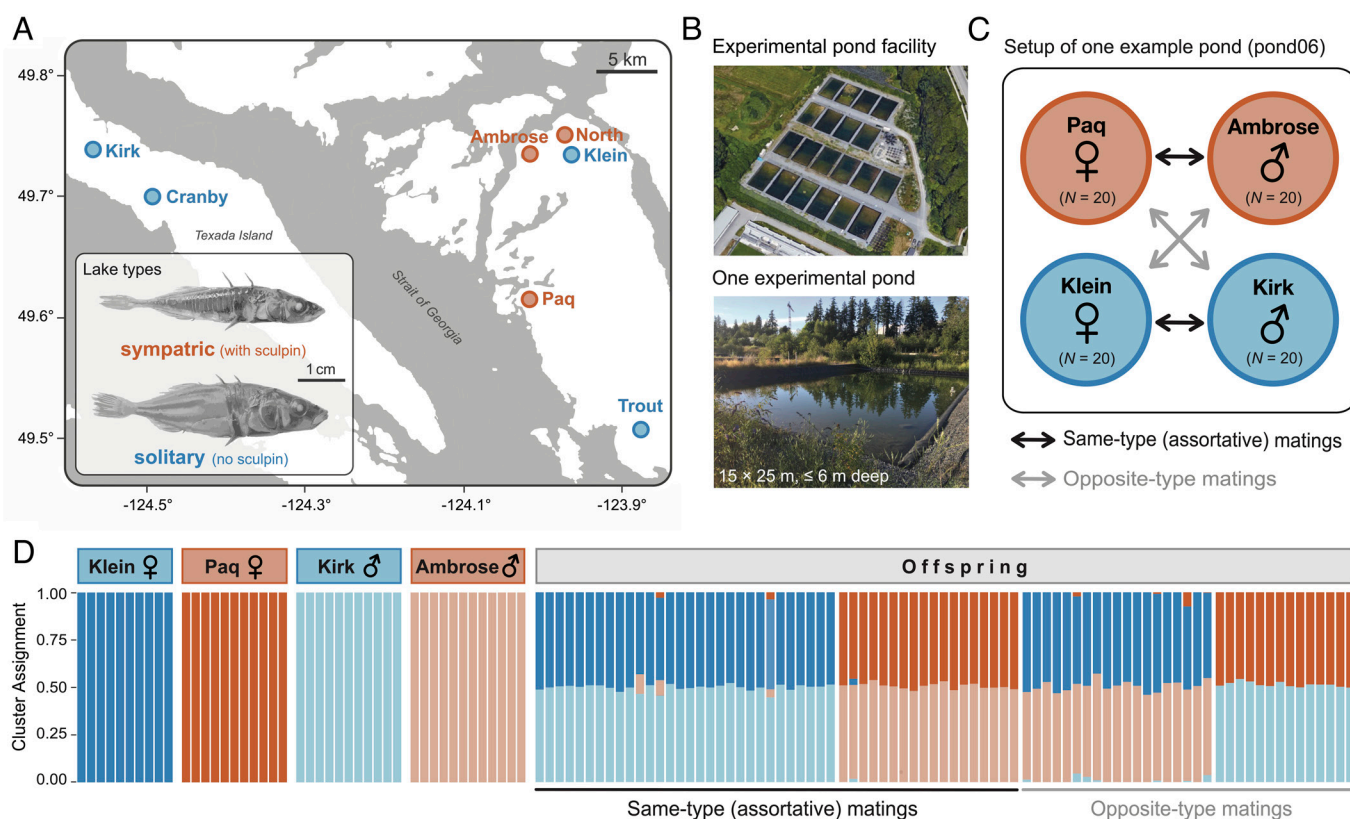


Fig. 1. Experimental system and design. (A) Map of the stickleback populations in British Columbia (Western Canada) used in the mating experiment. Here and in all figures, orange represents sympatric lakes where stickleback co-occur with sculpin, and blue indicates solitary lakes where stickleback occur without sculpin. The *Inset* shows representative photographs of sympatric and solitary stickleback. (B) Experimental facility used for the outdoor mating experiment. Experimental ponds were specifically built to contain typical stickleback habitats, including both deep open-water (limnetic) and shallow benthic zones. (C) Design of the mating experiment, illustrated by one example pond (pond06). Black arrows indicate possible matings between females and males of the same lake type (“assortative matings”), while gray arrows indicate possible matings between individuals of opposite lake types. (D) Genetic assignment of offspring from the pond experiment to parental populations, here exemplified by pond06. Individuals from each parental population (*Left*) served as a reference to assign admixed offspring (*Right*) to either same-type (assortative) or opposite-type mating events.

another, the seven experimental source lakes are otherwise geographically proximate and interspersed with respect to the presence or absence of sculpin, minimizing other possible systematic differences among them (Fig. 1*A*). To quantify prezygotic isolation, we experimentally simulated secondary contact in large, seminatural ponds designed to mimic natural stickleback habitats (Fig. 1*B*). Specifically, we captured adult stickleback at the start of their breeding season and stocked each of eight experimental ponds with males from one sympatric and one solitary lake, together with females from different sympatric and solitary lakes (with one exception; see *Materials and Methods* for details) (Fig. 1*C*). Males and females within a pond therefore never came from the same source lake, avoiding potential bias from within-lake mating preferences. We also ensured that each pond received fish from a unique combination of source lakes and that the numbers of sympatric versus solitary stickleback in each pond were standardized within each sex (16 to 20 fish per sex and source lake; see *SI Appendix, Table S1* for details). In total, this resulted in 573 stocked stickleback across the experiment.

Fish were allowed to mate freely in the ponds for approximately 2 mo until the end of their breeding season. We sampled entire nests throughout this time and free-swimming juvenile offspring at the end of the experiment (see *Materials and Methods* for details). For each nest, we recorded water depth and sampling date. We then extracted DNA from a single egg per clutch as well as from a random subset of the sampled free-swimming juveniles. Using genome-wide single nucleotide polymorphisms (SNPs) from reduced-representation genome sequencing, we identified offspring from between 30 to 81 distinct mating events per pond and determined the source populations of their parents (see *SI Appendix, Table S1* and *Materials and Methods* for details). This allowed us to quantify the frequency of mating events between females and males from the same lake type (i.e., both from sympatric lakes, or both from solitary lakes; hereafter simply assortative matings) and from opposite lake types (i.e., pairings between a solitary and a sympatric stickleback) (Fig. 1*D*). Positive assortative mating by lake type occurred when more than half of all matings (>0.5) happened between individuals from the same lake type.

Our experiment revealed that adaptation to sculpin presence/absence has resulted in some prezygotic isolation, but the observed isolation strength was variable and only moderate on average. That is, 253 of the 411 total mating events across all ponds (61%) occurred between males and females from the same lake type. However, the proportion was variable among replicates, ranging from 12 to 100%, with assortative matings exceeding opposite-type matings in 10 of 15 female populations (Table 1). This finding of partial and variable levels of prezygotic isolation among allopatric stickleback populations is consistent with their recent divergence.

We took advantage of this broad variation in isolation strength to test a key prediction of ecological speciation: that reproductive isolation increases with the degree of adaptive divergence between populations. In stickleback, assortative mating can arise from phenotype-driven mate choice, whereby females preferentially mate with males that are phenotypically similar (39–42). To help isolate the component reflecting adaptation to sculpin presence/absence from overall phenotypic variation, we identified the primary axis of body shape divergence in a larger independent set of six sympatric and five solitary populations not included in the pond experiment and then projected the experimental populations onto this axis. As expected under parallel adaptation to sculpin presence/absence, the experimental populations clustered along this axis by lake type yet also exhibited substantial variation (Fig. 2*A*). Strikingly, and consistent with ecological speciation, population variation along this axis was predictive of assortative mating: The frequency of assortative matings increased when females were more similar in shape to males of the same lake type than to males of the opposite lake type (Fig. 2*B*; $\chi^2 = 5.081$, $Df = 1$, $P = 0.024$). In contrast, when we considered overall shape variation among the experimental populations agnostic to sculpin presence/absence, no such association emerged ($\chi^2 = 0.002$, $Df = 1$, $P = 0.967$; *SI Appendix, Fig. S1A*; see also *SI Appendix, Supplementary Analysis 2* for a complementary approach yielding the same conclusions). Together, this suggests that variation in assortative mating among these recently evolved allopatric stickleback populations is explained by a specific component of

Table 1. Mating frequencies by population and lake type

Pond ID	Female population	Female population type	Male population same type	Male population opposite type	Same-type (assortative) matings	Opposite-type matings	Same-type (assortative) mating frequency
pond06	Klein	solitary	Kirk	Ambrose	30	19	0.612
pond06	Paq	sympatric	Ambrose	Kirk	18	14	0.563
pond07	North	sympatric	Ambrose	Cranby	20	10	0.667
pond07	Trout	solitary	Cranby	Ambrose	12	24	0.333
pond08	Ambrose	sympatric	North	Trout	6	4	0.600
pond08	Cranby	solitary	Trout	North	27	6	0.818
pond10	Paq	sympatric	Ambrose	Klein	10	16	0.385
pond10	Trout	solitary	Klein	Ambrose	6	0	1.000
pond11	Paq	sympatric	Ambrose	Cranby	8	8	0.500
pond11	Trout	solitary	Cranby	Ambrose	9	5	0.643
pond16	Kirk	solitary	Klein	Paq	34	11	0.756
pond16	North	sympatric	Paq	Klein	3	23	0.115
pond17	Cranby	solitary	Kirk	Paq	31	0	1.000
pond18	Ambrose	sympatric	North	Klein	3	5	0.375
pond18	Kirk	solitary	Klein	North	36	13	0.735

Number of matings between each female population and its two alternative male populations (i.e., males of the “same” versus “opposite” lake type) within the eight experimental ponds, as well as the resulting same-type (assortative) mating frequencies for each female population.

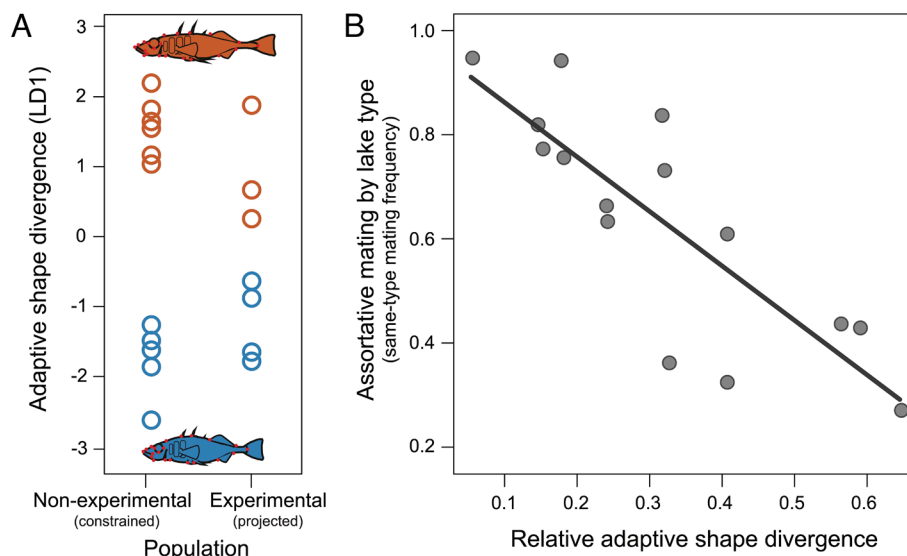


Fig. 2. Adaptive phenotypic divergence predicts assortative mating. (A) The principal axis of shape divergence from a linear discriminant analysis of body shape data from 11 nonexperimental populations with lake type (sympatric versus solitary) as the grouping variable. Fish from the seven experimental source populations projected onto LD1 separate by lake type but show variation along this axis. The *Inset* drawings illustrate typical body shape differences between sympatric (*Top*) and solitary (*Bottom*) stickleback populations, with red points indicating landmark locations used to capture shape variation. (B) Assortative mating by lake type, measured as the frequency of same-type matings for each female population, is predicted by its relative adaptive shape difference from the two alternative male populations available within each pond ($r = -0.826$, $P = 0.024$). The strength of assortative mating increases toward 1.0 when females are phenotypically close to males of their own lake type and distant from males of the opposite lake type (relative adaptive shape divergence < 0.5). Mating is close to random with respect to lake type (0.5) when females differ phenotypically to a similar extent from males of both lake types (relative adaptive shape divergence = 0.5). Dots and the regression line show the model fit with “pond” and “female population” included as random effects.

phenotypic divergence—namely, the component associated with sculpin presence/absence.

Following a similar rationale as in the phenotypic analyses, we also tested whether the strength in assortative mating is associated with the degree of adaptive genomic divergence. We obtained whole-genome sequence data for 86 total individuals from 24 sympatric and solitary lake populations, including 46 individual genomes from the seven source populations used in the pond experiment (*SI Appendix, Table S2*). To identify candidate genome regions repeatedly selected in response to sculpin presence/absence, we calculated genetic divergence in 20 kb windows across 2.66

million genome-wide SNPs between ten sympatric and seven solitary populations not included in the pond experiment. From this genome scan, we took the top 5% most divergent windows as candidate genome regions under repeated divergent selection in response to sculpin presence/absence (Fig. 3A). Indeed, divergence in these candidate adaptive genome regions was also greater between the experimental populations from opposite lake types than between those from the same lake type ($F_{st} = 0.46$ versus 0.36 , $P = 0.019$) (Fig. 3B), whereas this was not the case in the rest of the genome outside these candidate adaptive regions (same type: $F_{st} = 0.30$, opposite type: $F_{st} = 0.31$, $P = 0.779$). Importantly, the strength of parallel genomic

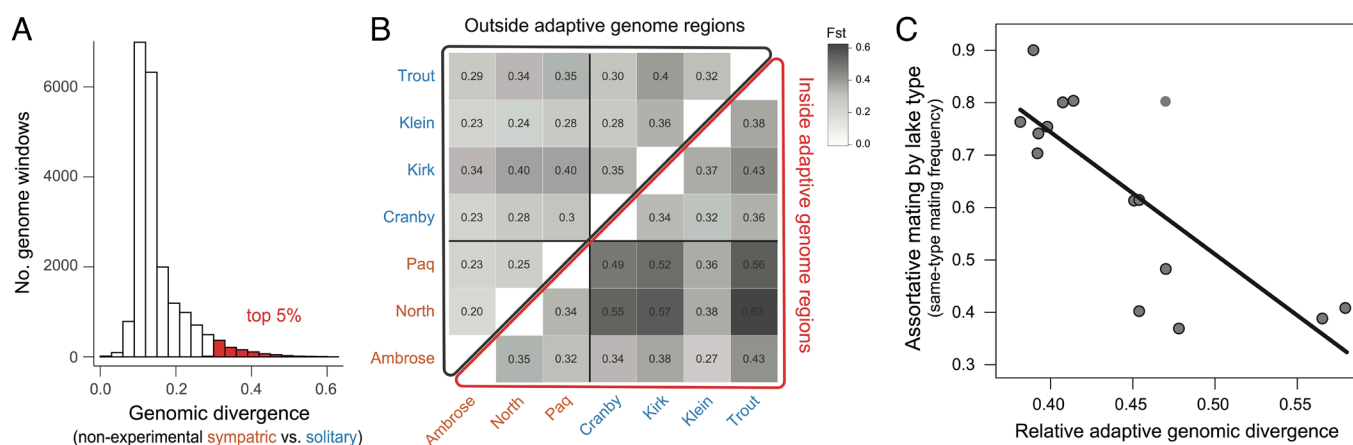


Fig. 3. Adaptive genomic divergence predicts assortative mating. (A) Genetic divergence in 20 kb genome windows between 20 stickleback individuals from each of 10 sympatric and 7 solitary populations that were not used in the mating experiment. The top 5% windows from this divergence scan were considered candidate genome regions under repeated divergent selection in response to sculpin presence/absence. (B) Pairwise genetic divergence (F_{st}) between the seven source populations used in the mating experiment. *Lower Right* portion (outlined in red) shows divergence within the candidate adaptive genome regions, quantified by mean F_{st} . *Upper Left* portion (outlined in black) shows mean F_{st} values across all SNPs located outside these adaptive genome regions. (C) Assortative mating by lake type, measured as the frequency of same-type matings for each female population, is predicted by its relative difference within adaptive genome regions from the two alternative male populations available for mating ($r = -0.764$, $P = 0.045$). The strength of assortative mating increases when females are genetically close to males of their own lake type within the adaptive genome regions and distant from males of the opposite lake type (relative adaptive genomic divergence < 0.5). Mating is close to random with respect to lake type (0.5) when females are similarly divergent from males of both lake types (relative adaptive genomic divergence = 0.5). Dots and the regression line show the model fit with “pond” and “female population” added as random effects.

divergence associated with sculpin presence/absence varied among the experimental populations, with F_{ST} within the adaptive genome regions between populations from opposite lake types ranging from 0.272 to 0.632 (Fig. 3B). As in the phenotypic analysis, this continuum of adaptive genomic divergence allowed us to assess quantitatively its relationship with assortative mating.

Consistent with the phenotypic results, we found that prezygotic isolation was strongest between sympatric and solitary populations with the greatest divergence within the adaptive genome regions. Specifically, the frequency of assortative matings was highest when females showed low adaptive genetic divergence from males of the same lake type and high divergence from males of the opposite lake type (Fig. 3C; $\chi^2 = 4.006$, $Df = 1$, $P = 0.045$). In contrast, genetic divergence calculated from SNP variation outside the adaptive genome regions, as well as across all genome-wide SNPs (SI Appendix, Fig. S2), showed no clear such association ($\chi^2 = 0.534$, $Df = 1$, $P = 0.465$ and $\chi^2 = 0.853$, $Df = 1$, $P = 0.356$; SI Appendix, Fig. S1 B and C). Together with the phenotypic findings, these genomic results strengthen the conclusion that it is specifically the divergence associated with sculpin presence/absence that underlies variation in a key component of recently evolved prezygotic isolation among these allopatric populations.

Assortative mating outcomes observed in our experiment could in part also reflect temporal or spatial isolation. We therefore additionally tested whether systematic differences in breeding depth or timing between sympatric and solitary populations might further contribute to assortative mating. However, we found no clear differences in breeding depth or timing between sympatric and solitary populations, nor any consistent spatial or temporal pattern in assortative mating (SI Appendix, Supplementary Analysis 3). Thus, spatial and temporal isolation are unlikely to be major contributors to the observed assortative mating by lake type, and we interpret our results as reflecting the outcome of phenotype-driven mate preferences.

Discussion

Our study provides experimental evidence that assortative mating—an important component of prezygotic isolation—can evolve as a by-product of adaptation to distinct ecological conditions during the initial allopatric phase of speciation. While ecological divergence has frequently been linked to the evolution of reproductive isolation, most documented cases involve speciation with gene flow (i.e., populations currently in sympatry or parapatry) and focus on later stages of the process (e.g., refs. 43–47). In contrast, direct tests of ecological divergence driving reproductive isolation in the early allopatric phase of speciation are scarce (48, 49). As a result, the role of ecological divergence in initiating speciation under geographic isolation has remained unclear, leading to a view that reproductive isolation in allopatry typically arises instead through neutral processes or selection unrelated to environmental differences (4, 7, 8, 50–53).

Our findings further underscore the challenge of identifying ecological divergence as a key driver of reproductive isolation in allopatry when multiple causes of divergence can potentially obscure this link. That is, we only detected assortative mating to be associated with phenotypic and genomic divergence when these were measured in relation to variation in a focal ecological selection pressure, whereas no clear association emerged when divergence was quantified without reference to this ecological difference. Thus, while our results point to an important role for ecological divergence in the early stages of allopatric speciation in stickleback, they also highlight the inherent difficulty of detecting ecological speciation under geographic isolation. More broadly, our findings

caution against viewing geographic isolation as an alternative to ecological speciation.

Our experiment revealed complete prezygotic isolation in two of fifteen female populations, highlighting the potential for rapid and complete ecological speciation in allopatry. However, prezygotic isolation was highly variable, and for most cases only weak to moderate, making it insufficient on its own to maintain population separation over multiple generations in the event of secondary contact. Indeed, even with very strong prezygotic isolation, coexistence would still require at least some postzygotic isolation (54). Thus, if geological conditions were to bring about secondary contact between these divergent populations in the near future, most would likely collapse by fusion. However, divergent ecological adaptation can also give rise to additional reproductive barriers upon secondary contact, including those causing reductions in hybrid fitness (55, 56). Thus, if selection also acted against hybrids, the combined effect of the postzygotic barriers and the reinforcement of prezygotic isolation could lead to well-isolated species (57, 58).

Indeed, such steps appear to have led to the formation of currently coexisting stickleback species pairs in several postglacial lakes in Western Canada. These benthic and limnetic ecotypes are thought to have originated repeatedly and in each case from a double invasion of the same lake, with the now-coexisting forms having diverged initially in allopatry. They persist as distinct species in the wild via divergent resource use, strong prezygotic isolation, and extrinsic postzygotic isolation (59, 60). Nonetheless, the ecological and reproductive isolation of these ecotypes is thought to be just above the critical threshold required for coexistence (61), and in at least one lake, previously isolated ecotypes collapsed into a hybrid swarm after an ecological disturbance, likely triggered by the introduction of exotic crayfish (62). Similar collapses of formerly reproductively isolated lineages have been observed in other systems where hybridization eroded species boundaries after ecological shifts (63–65). This suggests that while ecological isolation can evolve rapidly and form a strong barrier, its persistence remains vulnerable to environmental change. Strong intrinsic incompatibilities next to strong assortative mating might be necessary to prevent species collapse upon secondary contact (13, 54, 66). Yet, growing genomic evidence suggests that even closely related “good species” and deeply diverged lineages often continue to exchange genes while remaining distinct (67–70). Thus, the evolution of ecological differences in allopatry may be a key ingredient in both the initiation and long-term maintenance of reproductive isolation.

How can we explain such rapid evolution of prezygotic isolation between allopatric populations, as revealed by our experiment? A key factor appears to be that divergence in traits underlying assortative mating occurred via selection on abundant preexisting (standing) genetic variation. In threespine stickleback, past gene flow between freshwater and marine populations during interglacial cycles is thought to have resulted in a rich reservoir of standing genetic variation in the marine population. This variation is readily available for selection following recolonization of freshwater environments after glacial retreat (25, 71–74). As a result, much of the genetic variation underlying stickleback adaptation in the last ~12,000 y following the last glacial retreat is considerably older and was likely shaped by freshwater adaptation during previous interglacial periods (75, 76). Repeated selection on shared standing genetic variation can help explain the rapid parallel evolution of traits affecting mating preferences in ecologically similar lakes.

Besides facilitating the rapid evolution of prezygotic isolation via parallel adaptation, the availability of a rich pool of common genetic variation should also promote environment-dependent postzygotic

isolation if populations later meet in secondary contact (55, 56). This is because standing genetic variation provides immediate raw material for adaptation, allowing populations to diverge rapidly in response to ecological differences and thereby increasing the likelihood of maladaptive trait combinations in their hybrids (77). On the other hand, abundant standing variation should reduce the role of new mutations during adaptation, potentially slowing the accumulation of intrinsic incompatibilities, which often arise from positive selection on new mutations (78). Consistent with this idea, recently formed postglacial stickleback populations show little intrinsic isolation despite often substantial genetic divergence (55, 77, 79, 80). Thus, we speculate that the very conditions promoting rapid ecological isolation through parallel adaptive divergence in allopatry—namely, a wealth of standing genetic variation and repeated ecological differences—may at the same time dampen the evolution of intrinsic isolation, which typically arises from new mutations accumulating in geographically isolated populations (81, 82). Indeed, this idea could help resolve the conundrum of why species undergoing rapid ecological diversification and isolation are often not the ones evolving strong intrinsic barriers (83).

If ecological divergence plays a key role in the allopatric phase of speciation, which selective forces are likely to be involved? Intimate species interactions, such as host–parasite relationships or obligate mutualisms, are well known to facilitate rapid evolution of reproductive isolation (e.g., refs. 84–86). Our findings highlight that also indirect species interactions can be potent drivers of ecological speciation. Indeed, the strong effects of sculpin presence/absence on phenotypic and genomic divergence of stickleback arise primarily from the species' indirect trophic interactions, namely resource competition and shared predation (29). As we have shown here, variable ecological character displacement in threespine stickleback in response to prickly sculpin is closely associated with the strength of prezygotic reproductive isolation among stickleback evolving in the presence or absence of sculpin. Although ecological character displacement is widely documented in nature (e.g., refs. 87–92), demonstrating its direct link to reproductive isolation has so far proved challenging (17, 21, 93, 94). Given that ecologically distinct stickleback species that initially diverged in allopatry now coexist in several lakes, our findings raise the intriguing possibility that speciation can begin through differential interactions with more distantly related species in allopatry and later be strengthened by interactions among partially isolated sister populations upon secondary contact. To what extent nonintimate and indirect species interactions initiate and shape speciation more broadly remains an open empirical question.

Materials and Methods

Study System. This study investigates allopatric threespine stickleback populations from geographically isolated freshwater lakes in western British Columbia, Canada, focusing on populations from two ecological “lake types”: sympatric lakes, where stickleback coexist with prickly sculpin, and solitary lakes, where sculpin are absent. All study lakes harbor only one other native fish species, cutthroat trout (*Oncorhynchus clarkii*), which is a ubiquitous and abundant predator of both stickleback and sculpin (29). A subset of lakes additionally contain rainbow trout introduced in recent decades. Previous work has shown that sculpin interact with stickleback as resource competitors, indirectly via shared predators (especially trout), and rarely as direct predators (29). As a consequence, independently evolved stickleback populations from otherwise ecologically similar sympatric and solitary lakes exhibit parallel phenotypic and genomic divergence (26, 28–30), and sculpin presence/absence continues to explain much of this divergence even after accounting for abiotic variation among lakes (26, 29).

For the mating experiment, we selected stickleback from seven lakes: three sympatric lakes (Ambrose, North, Paq) and four solitary lakes (Cranby, Kirk, Klein, Trout) (29). These experimental source lakes were selected to be geographically proximate yet distinctly separated and interspersed with respect to lake type,

minimizing systematic environmental differences among them aside from the presence or absence of sculpin. Data from additional six to ten sympatric and five to seven solitary stickleback populations, dependent on the analysis, were included to obtain additional measurements for phenotypic and genomic analyses (see below). *SI Appendix, Fig. S3 and Table S2* provide an overview of all populations in this study.

Mating Experiment. To catch mature stickleback for the mating experiment, we set unbaited minnow traps within the breeding zone of each source lake at the beginning of the stickleback breeding season. We visually determined the sex and sexual maturity of fish and transported them in aerated coolers to the stickleback fish facility at the University of British Columbia (Vancouver, Canada). The fish were held, separated by sex, in indoor 100-L fish tanks for up to 1 wk before starting the outdoor experiment.

The mating experiment was conducted in eight seminatural outdoor ponds at the University of British Columbia in 2 y (two ponds in 2016 and six ponds in 2017). Each replicate pond measured 25 × 15 m, held 750,000 L of water, and was specifically constructed to mimic natural stickleback habitats, including a shallow littoral zone and a deep open-water zone (maximum depth of 6 m), typical vegetation, and natural stickleback prey (Fig. 1*B*) (95). Before fish were introduced into the ponds, they were acclimated gradually to outdoor pond water conditions. Each pond received stickleback from four different source lakes: males from one solitary and one sympatric lake, and females from another solitary and another sympatric lake (except pond17, for which we only had solitary females available) (*SI Appendix, Table S1*). By introducing fish of only one sex per source lake into a pond, mating between males and females from the same population was prevented, thereby controlling for potential within-lake mating preferences. Each pond received a unique but otherwise random combination of fish with respect to sex and source lake.

We stocked each pond with 16 to 20 fish from each of four source lakes, where the number of introduced males and females per lake type was always balanced. This resulted in a total of 66 to 80 fish per pond (except for pond17, which received 51 total fish from three source lakes) (*SI Appendix, Table S1*). These numbers were chosen to provide all males with enough space to establish a breeding territory (96, 97), and to ensure that every gravid female had mating opportunities with multiple nesting males from each lake type (a single stickleback male can simultaneously care for multiple egg clutches within its nest). Therefore, ponds were deliberately not fully saturated to avoid confounding estimates of mate choice with differences in competitive nesting ability of males, or female access to males from different populations. All stickleback males were introduced first into a pond, so that they could establish breeding territories and build nests, followed by all females a few days later.

Fish were allowed to mate freely for approximately 8 wk until the end of the breeding season (*SI Appendix, Table S1*). In 2016, the two ponds were left undisturbed and sampled once at the end (see below), whereas the six ponds from 2017 were systematically searched every 5 to 10 d for nests using snorkeling and scuba diving, with efforts to minimize disturbance. We recorded the water depth and date of each nest before carefully removing it intact and placing it in a plastic cup filled with pond water. Occasionally, we caught freshly hatched larvae near their nest. We inspected each nest for egg clutches, finding that 68% of the 168 nests across the six ponds in 2017 contained eggs (*SI Appendix, Fig. S4A*). Among nonempty nests, the average number of clutches was 2.5 (*SI Appendix, Fig. S4B*). The observation that some nests remained empty while males in other nests had accepted clutches from multiple females suggests that females were not limited by the availability of mating partners, as expected. We visually identified and separated multiple egg clutches within nests based on distinct egg clumps and developmental stages, noting that 17% of the 278 total egg clutches required further genetic assessment due to some uncertainty about whether they truly represented separate clutches and thus distinct mating events (see next *Materials and Methods* section).

Egg clutches were hatched at room temperature, after which the larvae were killed with buffered MS-222 prior to preservation in pure ethanol. Except for five clutches where no eggs developed (suggesting they were not fertilized and were thus excluded from further analysis), eggs hatched at a high frequency with no noticeable variation across ponds. This is consistent with findings from multiple previous laboratory crossing experiments, which suggest that intrinsic postzygotic incompatibilities between different stickleback populations are largely absent

(55, 77, 79, 80). To conclude the outdoor experiment after the stickleback breeding season (August/September), we used minnow trapping followed by rotenone poisoning to retrieve all remaining fish from within the ponds. These fish included wild adults introduced for the experiment, free-swimming juvenile offspring from overlooked nests or from rare mating events post-June in 2017, and surviving juvenile offspring fish from all mating events in the 2016 ponds. All fish were preserved in pure ethanol.

Genetic Assessment of Mating Events. We obtained DNA samples from one entire larva per clutch and from fin clips of a random subset of juvenile offspring caught at the end of the mating experiment. Additionally, we extracted DNA from 10 to 12 wild-caught, random individuals from every source lake for reference. DNA extraction followed a standard phenol-chloroform protocol with an RNase treatment step.

For genome-wide genotyping of all those fish, we constructed genotype-by-sequencing (GBS) libraries according to Elshire et al. (98). Briefly, genomic DNA was individually digested with the restriction enzyme Pst1 and ligated to barcoded adaptors (Pst1 cuts at approximately 314,000 sites within the 460 Mb threespine stickleback genome). We randomly pooled the resulting products in equal proportions into three GBS libraries, each of which was amplified in six independent PCR reactions to reduce the chance of PCR-induced errors during later variant calling. After pooling replicate PCR reactions per library, we reduced the number of represented genomic sites for sequencing by selecting sequence fragments in the size range of 350 to 550 base pairs from an agarose gel. To enhance sequence complexity and thus eventual read quality, 15% bacteriophage PhiX genomic DNA was added to each library before sequencing. Each library was then single-end sequenced to 100 base pairs on a separate HiSeq 2500 lane at Génome Québec in Montreal (Canada). At a later stage we performed additional reduced-representation sequencing of 26 randomly chosen juvenile offspring fish from pond10 and pond11, following the protocol described in refs. 99 and 100. This yielded a total of 453 offspring sequenced from across the eight experimental ponds (see *SI Appendix, Table S1* for details).

After parsing raw fastq reads by barcode, we removed the barcode sequence from each sequence read as well as entire reads containing any missing base calls ("N"s). We additionally trimmed remaining fragments of Illumina adaptors at the 3' ends of reads, retaining only sequences of a minimum length of 40 bp. We used *Novoalign* (Version 3.0.2) to align all reads to the latest stickleback reference genome assembly (101), employing read length-aware alignment thresholds and enforcing unique alignment. SAM files were converted to BAM format using *samtools*. We then used *angsd* (102) to generate one genotype likelihood file per pond, encompassing all offspring and respective reference individuals, by setting filters for genotypes (minimum quality score [minQ]: 20, minimum sequencing depth [geno_minDepth]: 3), sites (minimum map quality [minMapQ]: 30, maximum total sequencing depth [setMaxDepth]: 2.5-fold above average depth, minimum genotype calls [minInd] in $\geq 60\%$ of individuals), and SNPs [likelihood ratio test *P*-value [SNP_pval]: $< 1e-6$, minor allele frequency (MAF) [minMaf]: 0.05]. This process yielded genotype likelihoods for an average number of 18,857 SNPs per pond-wise genotype likelihood file (*SI Appendix, Table S3*). Using *NGSadmixture* (103), we estimated individual admixture proportions from genotype likelihoods, specifying $K = 4$ (except for pond17, where $K = 3$). This enabled clear assignment of 437 offspring to their parental populations (e.g., Fig. 1D; see *Dataset S1* for details).

We next aimed to exclude potential offspring resulting from the same mating event to avoid bias. This was necessary due to uncertainty in 17% of the sequenced eggs as to whether they originated from separate clutches within the same nest (see above), as well as the possibility that some of the sequenced juvenile offspring obtained at the end of the experiment were full siblings from the same mating event. To estimate pairwise relatedness between all offspring individuals per pond, we used *NGSremix* (104), a method specifically designed for inferring close relationships in admixed individuals from low-coverage data. This analysis considered the individuals' genomic ancestry proportions and estimated allele frequencies in the source populations, which proved essential to obtain realistic relatedness estimates because offspring in the mating experiment were admixed F1 individuals between relatively strongly differentiated source populations. With an expected kinship coefficient of 0.125 for half-siblings and 0.25 for full-siblings, we used a threshold of > 0.1875 to identify full-siblings. We applied this threshold to all eggs with recorded uncertainty about distinct

clutches, as well as to all sequenced free-swimming juvenile offspring collected at the end of the mating experiment. We thus identified 6% of all offspring pairs as originating from the same mating event and excluded one member of each pair from further analysis (*Dataset S2*). The resulting final dataset contained 30 to 81 unique mating events per pond, totaling 411 unique mating events (*SI Appendix, Table S1*).

Association between Phenotypic Divergence and Reproductive Isolation.

In stickleback, assortative mating can result from phenotype-based mate choice, with females typically preferring phenotypically similar males (40–42)—a pattern broadly observed across different animals (39, 105). We quantified assortative mating (by lake type) as the frequency of female matings with males of the same lake type, calculated separately for sympatric and solitary females within each pond. Specifically, we divided the number of matings between each female population and males of her own lake type by the total number of matings. The resulting index ranges from 0 to 1, with values above 0.5 indicating positive assortative mating by lake type—that is, when the majority of matings occurred between individuals of the same lake type. Notably, within the six ponds from 2017 where mating events were inferred from both genotyped nest clutches ($N = 263$) and free-swimming offspring fish captured at the end of the experiment ($N = 86$), assortative mating strength did not differ between these data sources ($t = -0.307$, $Df = 18.78$, $P = 0.762$; *SI Appendix, Table S1*). This supports the expectation that intrinsic postzygotic isolation is largely absent in this system and that possible extrinsic postzygotic isolation via viability selection did not bias our estimates of prezygotic isolation.

We used an additional set of nonexperimental populations to predict and better quantify the phenotypic response to sculpin presence/absence among the experimental populations, to then test its association with prezygotic isolation. That is, we made use of phenotypic data from additional six sympatric and five solitary stickleback populations alongside the seven experimental populations for this analysis (*SI Appendix, Fig. S3 and Table S2*). The phenotypic data consisted of body shape measurements from 21 fixed landmarks placed on lateral photographs of 16 to 21 adult wild-caught individuals from each of the 18 populations (total $N = 335$, with a balanced sex ratio within each population; see ref. 29 for details). Landmark data were Procrustes superimposed and size-dependent shape effects were removed by regressing the Procrustes coordinates of all individuals onto centroid size using the *R*-package *geomorph* (106). We then conducted a Linear Discriminant Analysis (LDA) using lake type (i.e., solitary and sympatric) as the grouping variable, based on individual shape residuals from the 11 populations not used in the experiment. This analysis produced the principal axis of shape divergence (LD1) in response to sculpin selection onto which we projected the shape residuals from individuals from the seven experimental source populations (Fig. 2A). This approach to define the sculpin-adaptive divergence axis using an external dataset and applying it to the experimental source populations is analogous to cross-validation in LDA, which helps avoid overfitting especially when the number of traits is large relative to sample size. Accordingly, the larger set of nonexperimental sympatric and solitary populations helped improve both the estimate of the adaptive divergence axis (by reducing sampling error) while maximizing independence. We note, however, that a complementary approach in which we used all populations to define LD1 produced similar results supporting the same conclusions (*SI Appendix, Supplementary Analysis 2*).

We used the variation along LD1 to quantify "adaptive shape divergence" between each female population and its two available male populations within a pond. That is, we calculated the absolute shape difference between each female population and the male population from i) the same lake type, and ii) the opposite lake type. To obtain "relative adaptive shape divergence" for each female population, we divided the "same-type distance" by the sum of both distances. Thus, a low value for relative adaptive shape divergence indicates that a female is phenotypically close to males from the same lake type and phenotypically distant from males of the opposite lake type. Notably, even when the calculated relative adaptive shape divergence was close to 0.5, indicating similar absolute LD1 distances between a female population and its two alternative male populations, they always fell on opposite sides of the female population along the adaptive divergence axis, reflecting the overall separation of sympatric and solitary populations (Fig. 2A). We then tested for a predicted negative association between relative adaptive shape divergence and assortative mating. That is, we expected the highest frequency of same-type matings (strongest assortative mating by

lake) when a female population was phenotypically similar to males of her own lake type and distant from males of the opposite lake type. This was done using a linear mixed model (LMM) with the "same-type mating frequency" of each female population as the dependent variable, "relative adaptive shape divergence" as the fixed effect, and "female population" and "pond" as random effects. The LMM was run using the *blmer()* function from the *blme* R-package (107), which reduces the risk of a boundary (singular) fit but tends to give conservative significance levels for fixed effects.

Whole-Genome Resequencing Dataset. We used whole genome data to first reassess the phylogenetic relationship of solitary and sympatric stickleback populations, and to then test whether adaptive genomic divergence predicts assortative mating. Our dataset comprised whole-genome resequencing data from a total of 110 stickleback individuals, including 94 newly generated genomes and 16 from a prior study (26). These individuals were sourced from 11 solitary (48 genomes) and 13 sympatric freshwater populations (38 genomes), whereby the seven source populations used in the pond experiment were each represented by seven genomes (except the North Lake population with only four genomes). Additionally, we included 24 genomes from two nearby marine sites to assess the phylogenomic relationships among all populations. [SI Appendix, Tables S2 and S6](#) provide an overview of all whole-genome-sequenced samples, along with sequencing details and summary statistics. Libraries for the newly generated genomes were prepared using the Nextera XT DNA Library Preparation Kit (Illumina). Whole-genome libraries were sent to Génomica Québec for paired-end sequencing of 150 bp reads on an Illumina HiSeq 4000.

Raw variants were then filtered using *bcftools* and *vcftools* to retain only biallelic autosomal SNPs, thus excluding variants on the known sex chromosomes chrY and chrXIX, as well as those on chrM (mitochondrial genome) and chrUn (unassembled scaffolds). We also removed genotypes of low quality (minGQ 20) and with read depths below 4, as well as sites with low quality (minQ 30), more than 30% missing genotype calls, a MAF below 0.05, unusually high average read depth (i.e., higher than two times the IQR from the upper quantile across all variants), and within repeat-masked genome regions. This yielded a final dataset of 2,900,846 high-quality SNPs for all 86 freshwater and 24 marine individuals combined (mean individual genotype depth = $13\times$, mean frequency of called genotypes per individual = 0.84; see [SI Appendix, Table S4](#) for details). This dataset was used to assess the phylogenomic relationship among these populations ([SI Appendix, Supplementary Analysis 1](#)). To examine the association between reproductive isolation and adaptive genomic divergence (see next *Materials and Methods* section), we excluded the 24 marine individuals from the dataset and refiltered for a MAF of 0.05, yielding 2,663,618 SNPs.

We next evaluated the difference in divergence within and outside these candidate adaptive genome regions in the populations used in the pond experiment. This was done by comparing mean divergence [i.e., mean F_{ST} ; (111)] across all SNPs inside and outside the predicted adaptive genome regions between experimental populations from different lake types (solitary-sympatric) versus the same lake type (sympatric-sympatric or solitary-solitary). To assess statistical significance, we performed permutation tests (112) in which we randomly reassigned lakes to the two lake types and recalculated mean genetic divergence between two types of populations (i.e., "opposite" or "same"). This was repeated 10,000 times to generate a null distribution and calculate a two-tailed P -value based on how often the absolute permuted differences exceeded the observed difference.

Finally, we quantified adaptive genomic divergence (mean F_{st} across all SNPs within the adaptive genome regions) between each female population and its two corresponding male populations within each pond. Following the approach used in the phenotypic analysis, we then calculated “relative adaptive genomic divergence” and tested whether it was negatively associated with the same-type mating frequency of each female population in an LMM. Analogous to the phenotypic analysis, a low value for relative genomic divergence indicates that a female population is genetically similar to males of the same lake type and genetically distant to males of the opposite lake type within the adaptive genome regions, which we predicted would result in a high frequency of assortative matings by lake type. Again using a similar rationale as in the phenotypic analysis, we assessed whether genomic population divergence agnostic to divergent sculpin selection explains assortative mating less effectively. We thus repeated the analysis but instead calculated mean F_{st} either by averaging across all SNPs outside the putative adaptive genome regions, or across all genome-wide SNPs. From these F_{st} values, we then again calculated relative genomic divergence for each female population in relation to its two corresponding male populations within each pond and tested its association with assortative mating in an LMM.

Statistical Analyses and Software. All analyses and plotting were conducted in R v.4.4.3 (113), unless noted otherwise. Because the main LMM fitted with *lmer()* from the *lme4* package resulted in singular fits, we used the *blmer()* function from the *blme* R-package (107). The *blmer()* function applies penalized likelihood to avoid zero variance estimates for random effects, resulting in more conservative (i.e., larger) *P*-values for fixed effects. *P*-values for fixed effects were obtained using the *Anova()* function from the *car* R-package (114), employing type III sums of squares. Model fits were visualized using the *visreg* R-package (115). [SI Appendix, Table S5](#) summarizes the outputs of the main statistical models used in this study.

Animal Care and Ethics. Collection (Permit Nos. MRSU16-229407 and SU17-258923) and transport (Permit Nos. VI16-225844 and VI16-260367) of stickleback were authorized by Fisheries and Oceans Canada (British Columbia). All procedures for the care and killing of stickleback were approved and monitored by the University of British Columbia Animal Care Committee under Animal Care Permit No. A16-0044.

Data, Materials, and Software Availability. Code and raw data used to generate the results of this study are available on Dryad under DOI: [10.5061/dryad.1ns1m96q](https://doi.org/10.5061/dryad.1ns1m96q) (116). Whole-genome and reduced-representation sequencing data have been

deposited in the NCBI Sequence Read Archive under project numbers [PRJNA1329683](#) (117) and [PRJNA1333495](#) (118), respectively.

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